CMP_SC-4650

HW4 - Detecting and Segmentation of Structures of Interest

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10 October 30

Abstract

In the medical field, being able to segment and identify different parts of microscopic images can be crucial for analyzing different diseases like cancer and parts of the cell. In this paper, blob detection algorithms will be used to identify blob-like shapes in these microscopic images, along with using k-means algorithms to segment different parts of the images based on parameters such as color.

Introduction

In the following paper, two images will be studied, one picture of cancer cells and the other of a kidney tumor. Through the use of image processing techniques, the formation of blobs in each image will be detected, as well as segmenting each image into different layers based on certain features within the image.

For the blob detection, Laplacian of Gaussian will first be applied to the image, which applies a gaussian filter to smooth out the noise and then the Laplacian operator. This isolates the edges within the images, preparing it for the blob detection. The gaussian filter size directly scales with the standard deviation of the filter (size $\approx 6\sigma$). This process takes advantage of the associative property of convolution for better efficiency, allowing for the gaussian filter to multiply with the Laplacian filter first, then doing convolution on the image using both at the

same time. Blobs are then able to be detected by locating the local maxima of the filtered image. The goal is to detect at which scale the nuclei is detected in the kidney tumor and which scale the nuclei is detected in the cancer.

For segmenting the image into various layers, the k-means algorithm is used, which creates a specific number of clusters chosen by the user and groups together pixels of closest similarity (in this case by color) of each cluster. This allows for parts of the image with vastly different values to be segmented and distinguished in both of the test images. The goal is to segment the nuclei

Experiments and Results

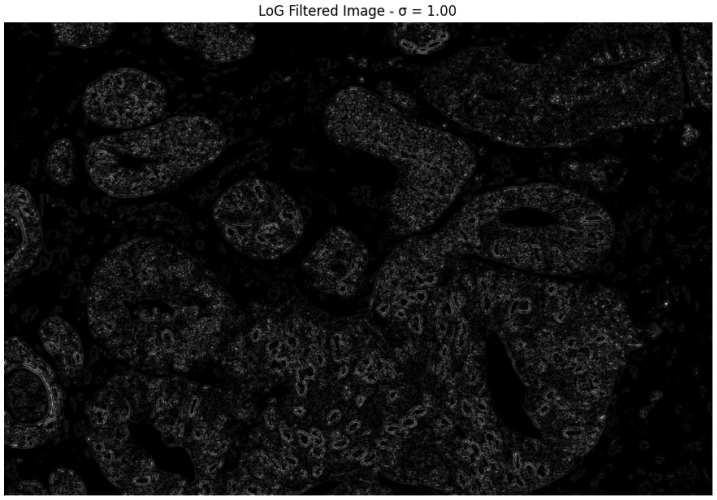


Figure 1. Cancer cell image with LoG filter applied with standard deviation of 1.

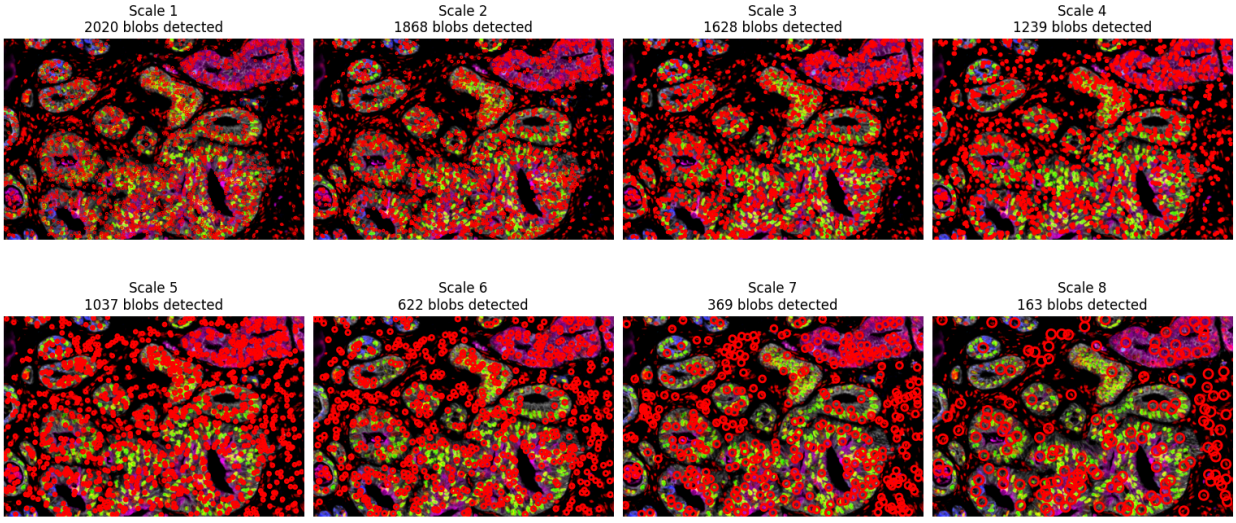


Figure 2. Blob detection for each scale for the cancer image.

When converting an image from RGB to grayscale, then applying blob detection to the image, the results turn out quite unsatisfactory. Blobs aren't very specific to any particular part of the image.

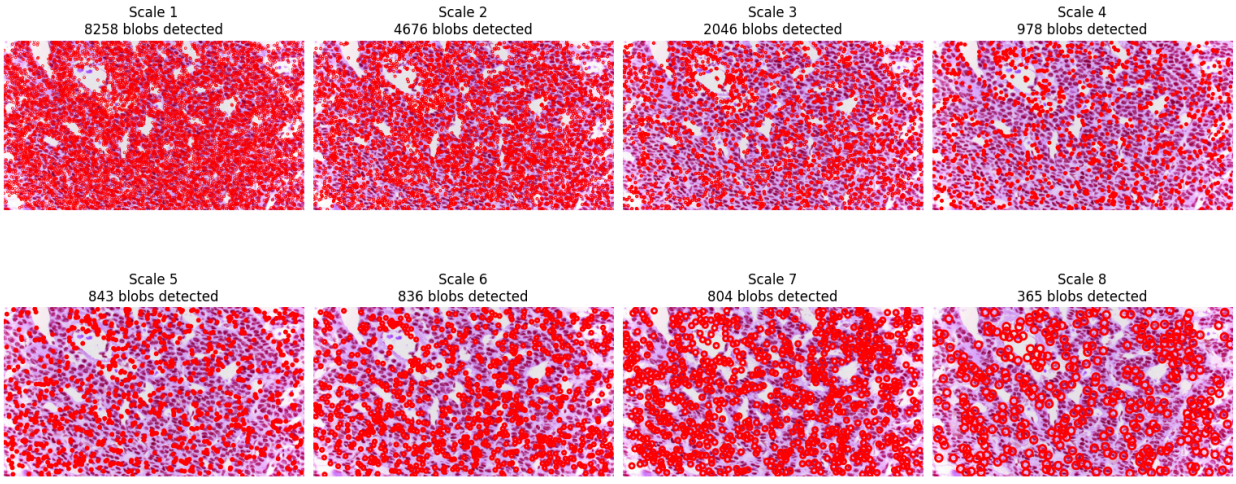


Figure 3. Blob detection for each scale for the kidney tumor image.

Similar results from figure 2 apply to this image. Blob centers and sizes are very accurate to the image.

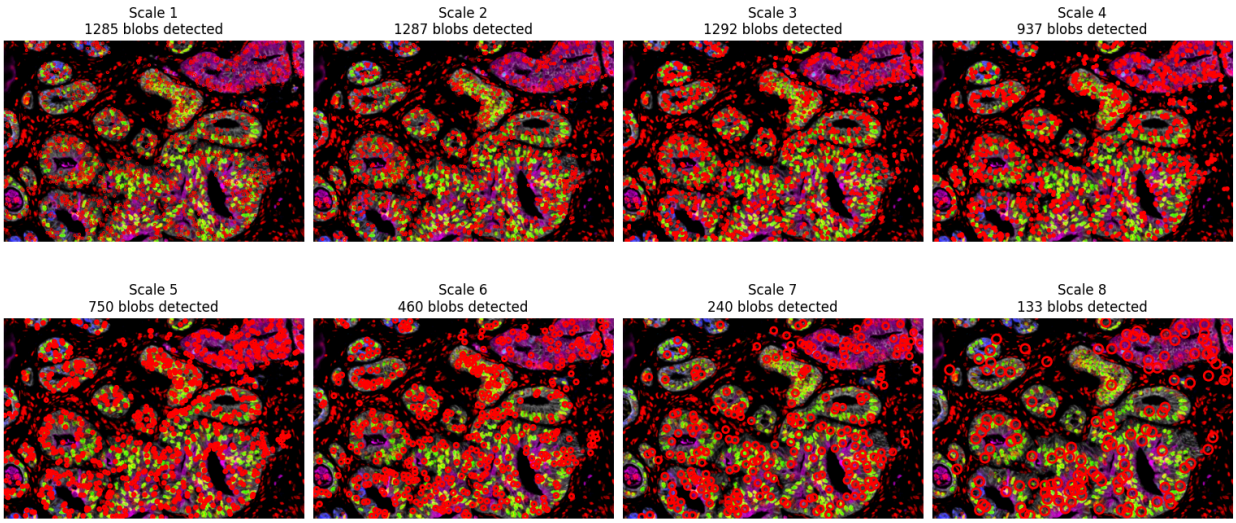


Figure 4. Blob detection for each scale for only the green channel of the cancer image.

When using only the green channel as the parameter for the blob detection, albeit not perfect, results in much better blobs surround the green parts of the image. The blobs most accurately being presented at scale 7.

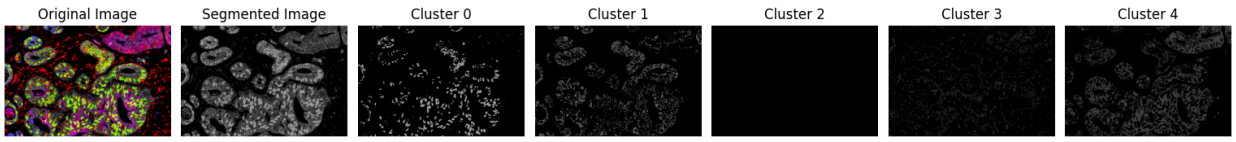


Figure 5. Segmented image of cancer cell using k-means.

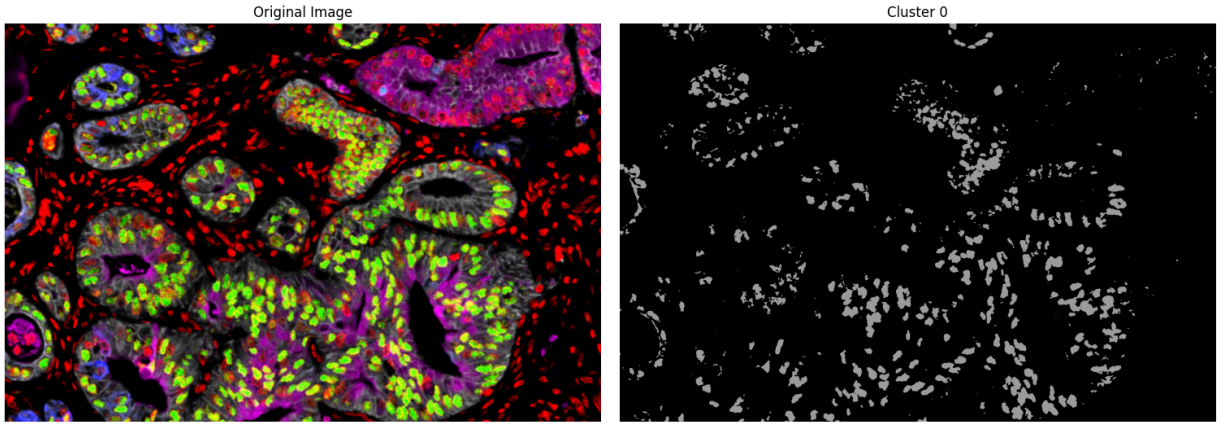


Figure 6. Isolated nuclei cluster from cancer cell image.

Seen from figures 5 and 6, the k-means algorithm was able to accurately segment the different colors in the image, showing how the nuclei can be completely isolated in figure 6.

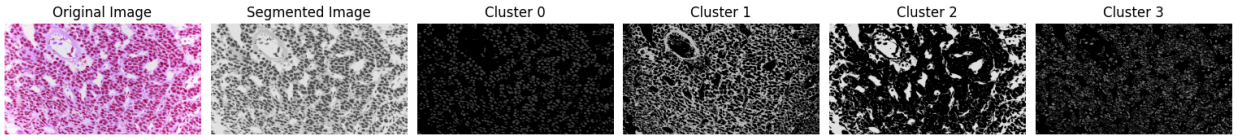


Figure 7. Segmented image of kidney tumor using k-means.

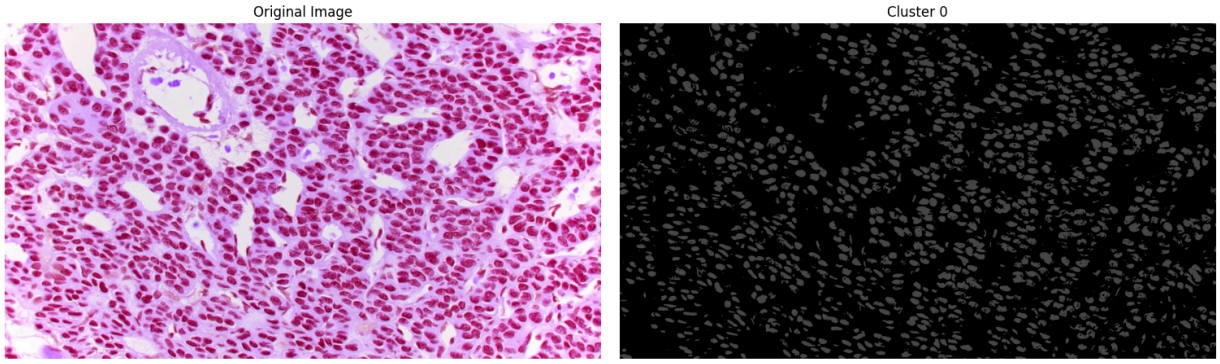


Figure 8. Isolated nuclei cluster from kidney tumor image

Similar results from figures 5 and 6 apply to figures 7 and 8.

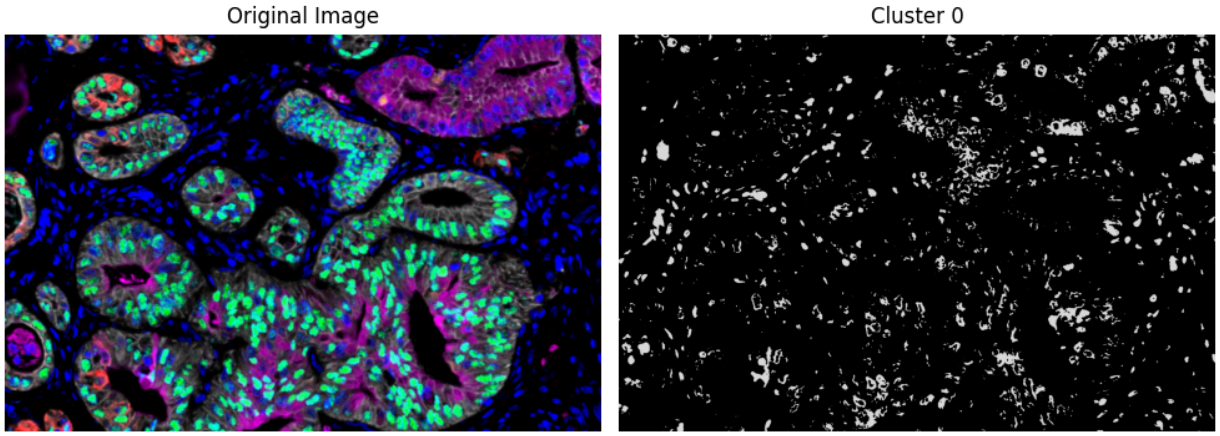


Figure 9. Isolated transparent blue cluster from cancer cell image (extra credit).

To achieve a clean separation of the blue blobs of the image, I used only the blue channel of the image into k-means, which allowed for this cluster to be segmented from the others.

Discussion and Conclusion

Through the process of creating the blob detection algorithm, there were many obstacles that were run into, and at the end the results were still not the best. The first major problem was that there were too many blobs being generated in an image. This was due to the fact that the algorithm was creating multiple blobs for one singular peak in the image, causing many very tightly packed circles in the image. This was minimized using a function that would remove blobs that were already too close to an accept blob. The distance needed for a blob to be removed was based on the average blob size within a given scale, therefore having more consistent results across scales. An issue with this strategy though, is that an accepted blob in a specific area can be slightly displaced from the actual center of the blob, therefore the blob for the real center would be removed. This is probably the cause for the misalignment of many of the blobs in the given images.

For more consistent results, the LoG kernel filters were created dynamically based on the window size and the variance. This made the filters more flexible as varying standard variations could be tested and it would automatically adjust accordingly. The kernel for each scale was also normalized so that it would only respond to local variations rather than global intensity variations.

The results from the k-means algorithm were quite good. The nuclei within the images were able to be segmented into their own cluster, and any specific color of the image could be particularly segmented by changing the color channel used as the input. The major issue with the k-means algorithm that was implemented is that the amount of clusters needed for each image has to be manually decided. For each image, I determined the k value by looking how many

unique colors there were. Luckily for the images tests, the colors were quite different, but for images where the colors don't contrast as much, this could be a major issue. A possible idea for improving this issue and automating the k value would be to test multiple k values, then determining the similarity of the range of pixels for each cluster. Increase the k value until the similarity of the pixels reaches a specific range, then you would have that k value to work off of. The downside to that is it would be very computationally expensive and would slow down the process immensely.

After applying these algorithms in this experiment, it is very clear to see how useful detecting blobs and segmenting images, especially in the medical field can be. It can be used to determine many properties of an image, such as amount of blobs, blob sizes, and area of specific attribute of an image.

Works Cited

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Lazebnik, S. *Achieving Scale Covariance • Goal: Independently Detect Corresponding Regions in Scaled Versions of the Same Image • Need Scale Selection Mechanism for Finding Characteristic Region Size That Is Covariant with the Image Transformation Blobs (and Scale Selection) Blob Detection in 2D*. www.cse.psu.edu/~rtc12/CSE586/lectures/featureExtractionPart2_6pp.pdf. Accessed 30 Oct. 2025.